AHMAD AYUB NU'MAN

AI & AUTOMATION ENGINEER

AI Agents Architecture • LLM Integrations • Business Workflow Automation • End-to-End Reliability

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PROFESSIONAL SUMMARY

AI & Automation Engineer with a rigorous full-stack software engineering foundation (2+ years) specializing in autonomous AI Agent design, Large Language Model (LLM API) integrations, and business process workflow automation. Proven expertise connecting backend services (JavaScript/TypeScript, Node.js, NestJS, Python) with relational databases (PostgreSQL, MySQL) and third-party systems via robust RESTful APIs and real-time webhooks. Track record includes architecting large-scale operational automation: automated batch document synthesis for SIMMACI (reducing 70% manual bottleneck across 270+ institutions), multi-branch retail POS & inventory management (Kiro ERP), and event-driven WhatsApp alert dispatching in Koneksi Santri. Certified in Spec-Driven Development (Kiro / Dicoding Academy), emphasizing rigorous error guardrails, deterministic failure handling, and comprehensive automated test suites (1,800+ test cases) to ensure AI Agents operate accurately, stably, and securely in production.

TECHNICAL SKILLS

AI Agents & LLMs:	AI Agent Architecture, LLM APIs (OpenAI, Anthropic, Gemini, Grok), Function / Tool Calling, n8n Workflow Automation, LangChain / LangGraph Concepts, Prompt Engineering, Agentic AI IDEs (Antigravity, Kiro, Codex)
Programming Languages:	JavaScript (ES6+), TypeScript, Python, SQL, PHP 8+, HTML5/CSS3
Backend & API Integration:	Node.js, NestJS, Express, RESTful API Integration, Webhook Processing, Inter-System Gateways (WhatsApp API, S3/MinIO), Microservices Architecture
Databases & Data Pipelines:	PostgreSQL 16, MySQL, Prisma ORM, Relational Schema Design, Query Optimization, ETL & Data Pipelines, Automated Reporting
Reliability & Testing:	Spec-Driven Development (SDD), Automated Testing (PHPUnit, Pest, Jest - 1,800+ Test Cases), Agent Error Handling & Guardrails, Log Analysis & Debugging, Performance Monitoring
DevOps, Git & Security:	Docker & Docker Compose, Linux/VPS Administration, Git / GitHub (1,940+ Commits), API Security Fundamentals (RBAC, Rate Limiting, OWASP, Token Sanitization)

PROFESSIONAL EXPERIENCE

Software & Automation Engineer (Lead Systems Architect & Product Lead) 2024 – Present

LP Ma'arif NU Cilacap • Cilacap, Central Java, Indonesia

- Architected and deployed operational workflow automation solutions and centralized data pipelines for 270+ schools, 2,000+ staff members, and 17,000+ students, cutting manual operational overhead by over 70%.
- Engineered secure RESTful APIs and webhook integration pipelines connecting external services: automated bulk document generation engines, isolated S3/MinIO cloud storage, and automated notification gateways.
- Engineered an automated institutional financial processing and reporting pipeline (Core Financial ERP) utilizing Domain-Driven Design (DDD) and state machines to ensure transaction ledger consistency and zero-discrepancy reconciliation.
- Integrated Agentic AI workflows (Antigravity, Kiro, Codex, Claude) into daily development for rapid business logic synthesis, automated test scenario formulation, and architectural refactoring.
- Enforced end-to-end system reliability and stability via 1,822 automated test cases (PHPUnit/Pest, 100% pass rate, 36,983 assertions), continuous debugging, and security guardrail audits (RBAC, API guard enforcement).
- Collaborated with cross-functional stakeholders to capture operational business requirements and translate them into scalable, high-efficiency technical solutions.

SELECTED AI AGENT, AUTOMATION & ENGINEERING PROJECTS

Enterprise POS & Multi-Warehouse Inventory (Kiro ERP) — Retail Automation Platform *React 19, NestJS, TypeScript, Prisma ORM, PostgreSQL, REST API, Apache ECharts*

Live Demo: inventory-pos-hazel.vercel.app | Repository: github.com/ayebe51/inventory-pos

- Engineered retail operational automation and multi-branch POS cashier systems with append-only inventory movement ledgers (WAC/FIFO) and live sales analytics reporting.
- Designed automated 3-way matching procurement workflows (PR/PO/Goods Receipt) and cashier reconciliations with sub-second transactional consistency.

SIMMACI — Large-Scale Process Automation & Batch Document Synthesis Platform *TypeScript, React 19, Laravel 12, PostgreSQL 16, REST API, Webhooks, PHPUnit/Pest*

Repository: github.com/ayebe51/simmaci

- Engineered an automated bulk document synthesis engine administering 270+ educational institutions, eliminating duplicate records and administrative turnaround latency.
- Implemented isolated cloud object storage integrations (S3/MinIO), webhook alert pipelines, and 1,822 automated test cases (100% pass rate, 36,983 assertions).

Koneksi Santri — Automated Ecosystem, WhatsApp Notification Gateway & Smart Gate *Next.js 14, Flutter (Dart), NestJS, PostgreSQL, Prisma, Webhook Gateway, Docker*

Live Demo: admin.koneksisantri.tech | Repository: github.com/ayebe51/koneksi-santri

- Integrated real-time WhatsApp webhook notification dispatches for guardian mobile monitoring of cashless student wallets and QR Smart Gate checkpoint authorisations.
- Developed periodic recurring tuition billing automations and multi-branch double-entry ledger balance reconciliations.

EDUCATION & QUALIFICATIONS

Bachelor of Islamic Economic Law (S.H.) Graduated 2019

STAI At-Tahdzib Jombang • Jombang, East Java, Indonesia

Strong logical reasoning, ethical integrity, and dedication to independent, continuous software engineering with active production systems.

CERTIFICATIONS & CREDENTIALS

Spec-Driven Development dengan Kiro April 2026

Dicoding Indonesia (Dicoding Academy) • ID: EYX4Q03J6PDL

Certified competence in Spec-Driven Development (SDD) using Kiro: comprehensive technical specification design, contract-first APIs, automated test coverage, and end-to-end requirement traceability.